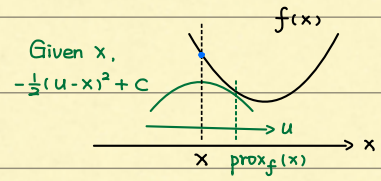
Proximal

$$\mathrm{prox}_f(x) = \arg\min_u f(u) + \frac{1}{2}\|u-x\|^2$$



后项梯度下降 $x^{k+1} = \mathrm{prox}_f(x^k) = x^k - \nabla f(x^{k+1})$

Moreau Envelope $e_f(x;c) = \frac{1}{2c}\min_u f(u) + \frac{1}{2}\|u-x\|^2$

1st Prox Thm：若 f 凸，则 $\mathrm{Prox}_f(x_0)$ 唯一点

2nd Prox Thm：

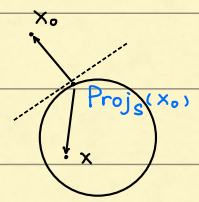
若 f 凸，$u \in \mathrm{Prox}_f(x^k) \iff x-u \in \partial f(u)$

Pf：$\min_u f(u) + \frac{1}{2}\|u-x\|^2$

Fermat's Opt：$0 \in \partial f(u) + u - x$ 故凸函数 $x^* = \mathrm{prox}_f(x^*)$

若 f 凸，则 $u_0 = \mathrm{Prox}_f(x_0)$

$\langle u_0 - x_0, u_0 - x\rangle \le f(x) - f(u_0) \quad \forall x$



Pf：$x_0 - u_0 \in \partial f(u_0)$，且 subgradient 定义

$f(x) \ge f(x_0) + \langle g, x-u_0\rangle$

$\Rightarrow$ Nonexpansiveness 非膨胀性

e.g. $f(x) = ax+b$，$\mathrm{prox}_f(x) = x-a$

$f(x) = \frac{1}{2}x^TAx + \langle b, x\rangle$，$\mathrm{prox}_f(x) = (A+I)^{-1}(x-b)$

Pf：可直接解 $u = x - \nabla f(u)$

按定义 $\min_u \frac{1}{2}u^TAu + \langle b, u\rangle + \frac{1}{2}\|u-x\|^2$

$Au + b + (u-x) = 0$，故 $u = (A+I)^{-1}(x-b)$

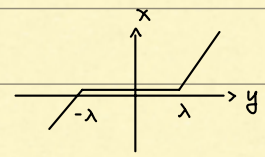
$I_C(x)$ proximal : Projection

$f(x) = I_C(x)$,

$$\text{prox}_f(x) = \arg\min \frac{1}{2}\|y-x\|_2^2 + I_C(x)$$

$$= \arg\min_{y \in C} \frac{1}{2}\|y-x\|_2^2 := \text{Proj}_C(x)$$

ℓ_1-norm proximal



$f(x) = \lambda\|x\|_1$,

$$\text{prox}_f(x) = \arg\min_x \frac{1}{2}\|y-x\|_2^2 + \lambda\|x\|_1$$ 即 LASSO 中的 A 取 I

$$= \text{Soft_thrd}_\lambda(x)$$

Pf : $f(x)$ 为 separable

$\min \frac{1}{2}(y-x)^2 + \lambda|x|$ $\quad x - y + \lambda\,\text{sgn}(x) = 0 \quad y = x + \lambda\,\text{sgn}(x)$

当 $y \geq \lambda$ 时, $x^* = y - \lambda$ $\quad$ 当 $y \leq \lambda$ 时, $x^* = y + \lambda$

当 $-\lambda < y < \lambda$ 时, $x^* = 0$

ℓ_0-norm proximal

$$\text{prox}_f(x) = \arg\min_x \frac{1}{2}\|y-x\|_2^2 + \lambda\|x\|_0 = \text{Hard_thrd}_{\sqrt{2\lambda}}(x)$$

Pf : $f(x)$ 为 separable

$\min \frac{1}{2}(y-x)^2 + \lambda I_{\neq 0}(x)$ $\quad x^*$ 只有 2 个选择 0 或 y

目标函数值 $\frac{1}{2}y^2$ 或 λ